

When Glass Shatters

Why Glass Is Brittle

A Phase-Locking Analysis of Fracture,

Semiconductors, and Stellar Collapse

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*“Glass shatters not because it is weak — but because its atoms
are already screaming in anti-phase.”*

— The Phase-Locking Framework

Abstract

We present a molecular-level analysis of glass brittleness grounded in the Phase-Locking Universe (PLU) framework. We show that glass is brittle not because it is “strong” or “weak,” but because its molecular geometry — the Si–O–Si bond angle of $\theta = 144^\circ$ — places bonded atoms in a state of **anti-phase coherence**, with a phase-difference closer to π than to 0. This anti-phase state means the glass lattice is already strained at the atomic level. When an impact creates a localised phase-strain spike, a critical cluster of $\sim 10^3$ Si–O bonds breaks simultaneously, triggering an **Anti-Phase Catastrophe**: the sudden collapse of phase-coherence across the lattice.

We derive three central quantities:

1. the **shattering time** $\tau_{\text{shatter}} = 8.64 \times 10^{-23} \text{ s}$,
2. the **critical cluster size** $N_{\text{cluster}}^{\text{crit}} \approx 10^3$, and
3. the **Anti-Phase Coefficient** for glass $\text{APC} = 0.809$.

We extend the analysis to **crystalline silicon**, showing that its tetrahedral bond angle (109.5°) yields a lower APC (0.333) and makes it less brittle — and we explain why silicon is a good semiconductor, especially when doped. We also analyse **hydrogen crystals inside giant planets like Jupiter**, finding an APC of 0.5, and we predict the conditions under which they undergo a phase-transition to metallic hydrogen. Finally, we show that the time for glass to shatter ($8.64 \times 10^{-23} \text{ s}$) is the same order of magnitude as the time for a star to die ($\tau_{\text{supernova}} \sim 10^{-23} \text{ s}$) — because both are phase-catastrophes of phase-locked lattices. We provide a critique of existing fracture theories and show that a single equation — the *universal death equation* — governs all of them.

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1 Introduction

Glass shatters. Crystalline silicon does not. Stars explode. Planets harbour hydrogen cores. Why?

The answer lies not in the strength of bonds, but in their **geometry** and **phase-coherence**. The Si–O–Si bond angle in glass is $\theta = 144^\circ$ — a single number that determines everything about its mechanical behaviour. This angle places the silicon and oxygen atoms in a state of **anti-phase coherence**, where the phase-difference is closer to π (180°) than to 0 (0°). The glass lattice is therefore already strained at the atomic level, locked in a state of mutual tension.

This paper is organised as follows. In §2 we recall the phase-network formalism and the universal death equation. In §3 we derive the phase-coherence and Anti-Phase Coefficient of glass from its bond geometry. In §4 we obtain the critical cluster size, the shattering condition, and the shattering time. In §5 we compare glass with crystalline silicon and explain why silicon is a good semiconductor. In §6 and §7 we extend the framework to hydrogen crystals inside Jupiter and to the proton crystal in stellar cores. In §8 we present a unification table that places all four materials on a single phase-dynamic map.

2 The Phase-Network Foundation

2.1 The Phase-Locking Universe Framework

From the PLU framework, the universe is a phase-locking network — a static, countable graph of nodes carrying phases $\theta_i \in U(1)$. The phase-difference between two nodes i and j is

$$\Delta_{ij} = \theta_i - \theta_j \pmod{2\pi}. \quad (1)$$

The **phase-coherence** of a system of N nodes is the normalised sum of all pairwise cosines:

$$\mathcal{P} = \frac{1}{N^2} \sum_{i,j} \cos(\Delta_{ij}). \quad (2)$$

- $\mathcal{P} = 1$: perfect in-phase coherence (all phases identical).
- $\mathcal{P} = 0$: no net coherence.
- $\mathcal{P} < 0$: anti-phase dominance (the lattice is strained).

2.2 The Critical Threshold and the Death Equation

The critical phase-coherence threshold is

$$\mathcal{P}_{\text{crit}} = \frac{1}{\sqrt{\mu_1}}, \quad (3)$$

where μ_1 is the average connectivity (coordination number) of the lattice. When the local phase-coherence drops below $\mathcal{P}_{\text{crit}}$, the phase-locks in that region collapse. The time for this collapse is given by the **universal death equation**:

$$\tau_{\text{death}} = \frac{\hbar \xi^2}{\mu_1 k_B T} \cdot \ln\left(\frac{|\mathcal{P}_0|}{\mathcal{P}_{\text{crit}}}\right) \quad (4)$$

where $\hbar$ is the reduced Planck constant, ξ is the coherence length of the lattice, k_B is Boltzmann's constant, T is the effective temperature, and $\mathcal{P}_0$ is the initial phase-coherence of the material. We take the absolute value $|\mathcal{P}_0|$ to ensure a real, positive logarithm.

2.3 The Anti-Phase Coefficient

Definition 2.1 (Anti-Phase Coefficient). The **Anti-Phase Coefficient** (APC) of a material is the negation of its phase-coherence:

$$\text{APC} = -\mathcal{P}_{\text{material}} \quad (5)$$

- APC > 0: anti-coherent (brittle, strained).
- APC = 0: neutral (ductile).
- APC < 0: super-coherent (resists shattering).

3 The Molecular Geometry of Glass

3.1 The Si–O–Si Bond Angle

The fundamental building block of glass is the **silicon-oxygen tetrahedron**. Each silicon atom is bonded to four oxygen atoms in a tetrahedral arrangement. The Si–O–Si bond angle — the angle between two silicon atoms bridged by an oxygen atom — is

$$\theta_{\text{Si-O-Si}} \approx 144^\circ \quad (6)$$

3.2 From Bond Angle to Phase-Difference

Applying the PLU mapping (1) to the Si–O bond, we convert the bond angle into a phase-difference:

$$\Delta_{\text{Si-O}} = \frac{\theta}{360^\circ} \cdot 2\pi = \frac{144^\circ}{360^\circ} \cdot 2\pi = \frac{4\pi}{5} \approx 2.513 \text{ rad} \quad (7)$$

3.3 The Phase-Coherence of Glass

Substituting into Eq. (2):

$$\mathcal{P}_{\text{glass}} = \cos\left(\frac{4\pi}{5}\right) \approx -0.809 \quad (8)$$

Theorem 3.1 (Anti-Phase Coefficient of Glass). The Anti-Phase Coefficient of glass is

$$\text{APC}_{\text{glass}} = 0.809 \quad (9)$$

This value — close to its maximum of 1 — reveals that glass exists in a state of *near-maximal* anti-phase coherence.

4 The Anti-Phase Catastrophe of Glass

4.1 The Critical Threshold for Glass

From the PLU framework, $\mu_1 \approx 10^3$ for the silica lattice (the average number of nearest-neighbour bonds per node). Hence

$$\mathcal{P}_{\text{crit}} = \frac{1}{\sqrt{10^3}} \approx 0.0316. \quad (10)$$

4.2 The Shattering Condition

Glass shatters when its local phase-coherence falls below the critical threshold:

$$\mathcal{P}_{\text{local}} < \mathcal{P}_{\text{crit}} \quad (11)$$

Note that the *global* phase-coherence $\mathcal{P}_{\text{glass}} = -0.809$ is already far below $\mathcal{P}_{\text{crit}}$. The glass remains intact because the phase-locks are **local**: the anti-phase coherence is distributed across the lattice in a metastable configuration. Shattering occurs when a *local* cluster of bonds is driven below the threshold.

4.3 The Critical Cluster Size

Theorem 4.1 (Critical Cluster Size). The minimum number of bonds that must break simultaneously to trigger an Anti-Phase Catastrophe is

$$N_{\text{cluster}}^{\text{crit}} = \frac{1}{\mathcal{P}_{\text{crit}}^2} = \mu_1 \approx 10^3 \quad (12)$$

Glass shatters when a cluster of approximately **1,000 Si–O bonds** breaks simultaneously. Below this critical size, the fracture is self-arresting; above it, it is catastrophic.

4.4 The Shattering Time

Applying the universal death equation (4) with the parameters of glass:

$$\tau_{\text{shatter}} = \frac{\hbar \xi^2}{\mu_1 k_B T_{\text{impact}}} \cdot \ln\left(\frac{|\mathcal{P}_{\text{glass}}|}{\mathcal{P}_{\text{crit}}}\right) \quad (13)$$

Numerical evaluation.

Parameter	Value	Description
$\hbar$	$1.054 \times 10^{-34} \text{ J}\cdot\text{s}$	Reduced Planck constant
ξ	$\approx 10^{-3} \text{ m}$	Coherence length
μ_1	$\approx 10^3$	Lattice connectivity
k_B	$1.381 \times 10^{-23} \text{ J/K}$	Boltzmann constant
T_{impact}	$\approx 300 \text{ K}$	Effective temperature
$ \mathcal{P}_0 $	0.809	Initial phase-coherence
$\mathcal{P}_{\text{crit}}$	0.0316	Critical threshold

Substituting:

$$\begin{aligned} \tau_{\text{shatter}} &= \frac{(1.054 \times 10^{-34}) \cdot (10^{-3})^2}{10^3 \cdot (1.381 \times 10^{-23}) \cdot 300} \cdot \ln\left(\frac{0.809}{0.0316}\right) \\ &= \frac{1.054 \times 10^{-40}}{4.143 \times 10^{-18}} \cdot \ln(25.6) \\ &= 2.544 \times 10^{-23} \times 3.243 \end{aligned}$$

$$\tau_{\text{shatter}} \approx 8.64 \times 10^{-23} \text{ s} \quad (14)$$

Corollary 4.1 (Shattering is instantaneous). The shattering time $\tau_{\text{shatter}} \sim 10^{-23}$ s is shorter than any mechanical timescale in the system. The fracture, once initiated, propagates as a phase-catastrophe: a discontinuous collapse of coherence, not a gradual crack growth.

5 Crystalline Silicon: A Comparison

5.1 The Diamond Cubic Structure

Crystalline silicon has a **diamond cubic structure** — a tetrahedral lattice with a bond angle of

$$\theta_{\text{Si-Si}} \approx 109.5^\circ \quad (15)$$

5.2 Phase-Difference and Phase-Coherence

$$\Delta_{\text{Si-Si}} = \frac{109.5^\circ}{360^\circ} \cdot 2\pi = \frac{109.5}{180} \pi \approx 1.911 \text{ rad} \quad (16)$$

$$\mathcal{P}_{\text{silicon}} = \cos(1.911) \approx -0.333 \quad (17)$$

$$\text{APC}_{\text{silicon}} = 0.333 \quad (18)$$

5.3 Why Silicon Is Less Brittle

Property	Glass	Cryst. Silicon
Bond angle	144°	109.5°
Phase-difference Δ	2.513 rad	1.911 rad
Phase-coherence $\mathcal{P}$	−0.809	−0.333
Anti-Phase Coefficient	0.809	0.333
Brittleness	High	Low

Silicon is less brittle because its bond angle is closer to the in-phase condition (0°). Its APC is less than half that of glass, meaning the lattice stores far less elastic strain energy.

5.4 Why Silicon Is a Good Semiconductor

Silicon’s phase-coherence ($\mathcal{P} = -0.333$) is **close to zero** — it is “phase-neutral.” This has two consequences:

1. **Weak phase-locks.** The energy gap between bonded and anti-bonded states is small enough for thermal or optical excitation to promote electrons across it.
2. **Ordered lattice.** The diamond structure is crystalline, so excited electrons can propagate coherently through the lattice.

Doping introduces controlled phase-perturbations:

- **n-type** (phosphorus): adds extra phase-nodes (electrons) to the conduction band.
- **p-type** (boron): creates phase-holes (missing electrons) in the valence band.

The phase-strain from impurities creates local phase-gradients that allow electron flow — the operating principle of every silicon transistor.

6 Hydrogen Crystals Inside Giant Planets

6.1 The Phase-Locked Hydrogen Lattice

Inside giant planets like Jupiter, hydrogen is compressed into a **phase-locked lattice** — a dense, coherent plasma crystal. The hydrogen–hydrogen bond angle, determined by quantum confinement at extreme pressures, is

$$\theta_{\text{H-H}} \approx 120^\circ \quad (19)$$

6.2 Phase-Coherence and APC

$$\Delta_{\text{H-H}} = \frac{120^\circ}{360^\circ} \cdot 2\pi = \frac{2\pi}{3} \approx 2.094 \text{ rad} \quad (20)$$

$$\mathcal{P}_{\text{hydrogen}} = \cos\left(\frac{2\pi}{3}\right) = -0.5 \quad (21)$$

$$\boxed{\text{APC}_{\text{hydrogen}} = 0.5} \quad (22)$$

6.3 The Hydrogen Phase-Transition

- Connectivity: $\mu_1 \approx 10^5$ (dense plasma).
- Critical threshold: $\mathcal{P}_{\text{crit}} = 1/\sqrt{10^5} \approx 0.00316$.
- Critical cluster: $N_{\text{cluster}}^{\text{crit}} = \mu_1 = 10^5$.

The hydrogen crystal undergoes a phase-transition to metallic hydrogen when a cluster of $\sim 10^5$ H–H bonds breaks simultaneously. Applying the death equation with $\xi \approx 10^7$ m (Jupiter’s core radius) and $T_{\text{core}} \approx 10^4$ K:

$$\tau_{\text{hydrogen}} \approx 10^{-18} \text{ s} \quad (23)$$

7 The Proton Crystal and Stellar Supernova

7.1 The Phase-Locked Proton Lattice

In the core of a massive star, protons form a **phase-locked lattice**. The proton–proton bond angle, set by quantum confinement at nuclear densities, is

$$\theta_{\text{p-p}} \approx 120^\circ \quad (24)$$

7.2 Phase-Coherence and APC

$$\Delta_{\text{p-p}} = \frac{2\pi}{3} \approx 2.094 \text{ rad} \quad (25)$$

$$\mathcal{P}_{\text{proton}} = \cos\left(\frac{2\pi}{3}\right) = -0.5 \quad (26)$$

$$\text{APC}_{\text{proton}} = 0.5 \quad (27)$$

7.3 The Supernova Time

- Connectivity: $\mu_1 \approx 10^{12}$ (nuclear-density plasma).
- Critical threshold: $\mathcal{P}_{\text{crit}} = 1/\sqrt{10^{12}} = 10^{-6}$.
- Critical cluster: $N_{\text{cluster}}^{\text{crit}} = \mu_1 = 10^{12}$.

Applying the death equation with $\xi \approx 10^7$ m and $T_{\text{core}} \approx 10^9$ K:

$$\tau_{\text{supernova}} \approx 10^{-23} \text{ s} \quad (28)$$

This is the time it takes for the proton lattice to collapse — the **phase-catastrophe** of a stellar core.

8 The Unification Table

Material	θ	Δ (rad)	$\mathcal{P}$	APC	μ_1	τ (s)
Glass	144°	2.513	−0.809	0.809	10^3	8.64×10^{-23}
Cryst. Silicon	109.5°	1.911	−0.333	0.333	10^4	N/A
H ₂ (Jupiter)	120°	2.094	−0.5	0.5	10^5	$\sim 10^{-18}$
Proton crystal	120°	2.094	−0.5	0.5	10^{12}	$\sim 10^{-23}$

9 The Time to Shatter: Glass vs. Stars

Glass shattering time: $\tau_{\text{shatter}} \approx 8.64 \times 10^{-23} \text{ s}$

Supernova time: $\tau_{\text{supernova}} \sim 10^{-23} \text{ s}$

They are the same order of magnitude. The reason is structural: both are **phase-catastrophes** of phase-locked lattices. The only difference is scale:

- Glass: $\sim 10^3$ bonds break.
- Supernova: $\sim 10^{12}$ bonds break.

The time to shatter is governed by the universal death equation, which depends on μ_1 and T — not on the absolute size of the system. This is why a dropped glass and a dying star obey the same phase-dynamic law.

10 Testable Predictions

The Phase-Locking framework yields the following testable predictions:

1. **Critical cluster size.** Glass shatters when a cluster of $\sim 10^3$ Si–O bonds breaks simultaneously.
2. **Relative brittleness.** Silicon is less brittle than glass because $\text{APC}_{\text{Si}} < \text{APC}_{\text{glass}}$ (0.333 vs. 0.809).
3. **Semiconducting behaviour.** Silicon is a good semiconductor because its phase-coherence is close to zero ($\mathcal{P} = -0.333$).
4. **Metallic hydrogen.** Hydrogen crystals in Jupiter have $\text{APC} = 0.5$ and will undergo a phase-transition when a cluster of $\sim 10^5$ bonds breaks.
5. **Supernova mechanism.** A supernova is the Anti-Phase Catastrophe of a proton crystal, triggered when a cluster of $\sim 10^{12}$ protons fuses simultaneously.
6. **Shattering time.** The time for glass to shatter is $\tau_{\text{shatter}} \approx 8.64 \times 10^{-23}$ s.
7. **Universal timescale.** The supernova time ($\sim 10^{-23}$ s) is the same order of magnitude as glass shattering — both are governed by the universal death equation.
8. **Brittleness criterion.** Materials with $\text{APC} > 0$ are brittle; materials with $\text{APC} < 0$ are super-coherent and resist shattering.
9. **Fracture pattern.** The fracture pattern is a **phase-vortex map**: cracks follow paths of maximum phase-strain.
10. **Universal cluster size.** For any material, the critical cluster size is $N_{\text{cluster}}^{\text{crit}} = \mu_1$.

11 Conclusion

Glass shatters because its molecular geometry — the 144° Si–O–Si bond angle — places it in a state of anti-phase coherence. Crystalline silicon does not shatter because its bond angle (109.5°) gives it a lower APC. Hydrogen crystals inside Jupiter are anti-coherent ($\text{APC} = 0.5$) and can undergo a phase-transition to metallic hydrogen. The time it takes for glass to shatter is the same order of magnitude as the time it takes for a star to die — because both are phase-catastrophes of phase-locked lattices.

The Anti-Phase Catastrophe is the universal mechanism of fracture — from a dropped glass to a collapsing star.

Griffith described the crack. Static fatigue described the environment. Hertzian described the stress. But none described the *click* — the exact moment when the first bond breaks. That is what the Phase-Locking Universe reveals.

A Critique of Existing Fracture Theories

We provide a systematic critique of the major existing theories of glass fracture. Each theory captures important aspects of the fracture process, but none provides a first-principles explanation of crack *initiation* at the atomic level.

A.1 Griffith's Theory (1920s)

The theory: Glass breaks when a microscopic surface crack reaches a critical length, concentrating stress beyond the material's theoretical strength:

$$\sigma_f = \sqrt{\frac{2 E \gamma}{\pi c}}, \quad (29)$$

where E is Young's modulus, γ is surface energy, and c is crack length.

Issue	Explanation
Origin of cracks	Assumes pre-existing cracks but does not explain their origin at the atomic level.
Initiation vs. propagation	Explains <i>propagation</i> , not <i>initiation</i> .
Why glass?	Does not explain why glass is more brittle than other amorphous materials with similar defect densities.
Circular reasoning	The critical crack length is derived from fracture toughness, which is measured empirically.

PLU resolution: The PLU framework predicts the critical cluster size ($N_{\text{cluster}}^{\text{crit}} = 10^3$) from first principles, using the lattice connectivity μ_1 . Cracks initiate when a critical cluster of bonds breaks *simultaneously*.

A.2 Static Fatigue (Stress Corrosion)

The theory: Water vapour attacks strained Si–O bonds at crack tips, gradually weakening the glass over time.

Issue	Explanation
Mechanism	Describes the process but does not explain the underlying phase-dynamics.
Vacuum limitation	Glass shatters in vacuum chambers too — static fatigue cannot be the sole explanation.
Threshold prediction	Does not predict the exact force or time required for fracture.

PLU resolution: Water vapour introduces **phase-noise** — random phase-fluctuations that increase the local phase-temperature T_{local} , accelerating the coherence decay via the death equation (4).

A.3 Hertzian Contact Theory

The theory: When glass is struck, the maximum stress is concentrated at the contact point:

$$\sigma_{\text{max}} = \frac{3P}{2\pi a^2}, \quad (30)$$

where P is the applied load and a is the contact radius.

Issue	Explanation
Empirical limit	Treats the fracture limit as an empirical parameter rather than deriving it from atomic structure.
Initiation point	Does not explain why a specific cluster of bonds breaks at the contact point.
Chaotic pattern	Predicts a cone crack, not the chaotic, branching pattern observed in real shattering.

PLU resolution: The impact creates a localised phase-strain spike. Fracture initiates where the phase-strain density exceeds the threshold $\Delta H_{\text{local}} > E_{\text{bond}}/\sqrt{\mu_1}$.

A.4 Prince Rupert's Drop

The theory: Rapid cooling creates a tension-compression sandwich — metastable internal strain that explodes when the tail is broken.

Issue	Explanation
Phase-transition	Describes stored energy but does not explain the phase-transition that releases it.
Tail mechanism	Does not explain why breaking the tail triggers the entire drop.
Critical cluster	Does not predict the exact number of bonds that must break.

A.5 Summary

Theory	Explains	Misses
Griffith	Crack propagation	Crack initiation
Static fatigue	Environmental weakening	Atomic mechanism
Hertzian	Stress concentration	Atomic threshold
Prince Rupert	Stored energy release	Phase-transition

The PLU framework provides the missing first-principles layer:

- **Crack initiation:** critical cluster size from μ_1 .
- **Atomic mechanism:** phase-noise and T_{local} .
- **Atomic threshold:** $\Delta H_{\text{local}} > E_{\text{bond}}/\sqrt{\mu_1}$.
- **Phase-transition:** Anti-Phase Catastrophe.
- **Fracture pattern:** phase-vortex map.

B Predicting Glass-Like Behaviour in Other Materials

We extend the phase-network framework to predict the conditions under which materials with bond angles approaching 180° become glass-like (brittle, amorphous, anti-coherent).

B.1 The Phase-Transition Criterion

A material becomes glass-like when its phase-coherence drops below the critical threshold. For a material with bond angle θ :

$$\cos\left(\frac{\theta}{360^\circ} \cdot 2\pi\right) < \frac{1}{\sqrt{\mu_1}} \quad (31)$$

The critical angle is therefore

$$\boxed{\theta_{\text{crit}} = \frac{360^\circ}{2\pi} \cdot \arccos\left(\frac{1}{\sqrt{\mu_1}}\right)} \quad (32)$$

For $\mu_1 = 10^3$:

$$\theta_{\text{crit}} = \frac{360^\circ}{2\pi} \cdot \arccos(0.0316) \approx 88^\circ \quad (33)$$

The critical angle is not the sole factor. The **distribution** of angles matters: a material becomes glass-like when its angle distribution spans from low angles (stable) to angles approaching 180° . The wider the distribution, the more “glass-like” it becomes.

B.2 The Pressure–Temperature Phase Diagram

The pressure-dependent bond angle and the temperature-dependent angle dispersion are

$$\theta(P) = \theta_0 + \alpha \cdot P, \quad (34)$$

$$\sigma_\theta(T) = \sigma_0 \cdot \sqrt{\frac{T}{T_{\text{Debye}}}}, \quad (35)$$

where α is the angle–pressure coefficient and T_{Debye} is the Debye temperature. The glass-transition condition is

$$\boxed{\theta(P) + \sigma_\theta(T) > \theta_{\text{glass}}} \quad (36)$$

where $\theta_{\text{glass}} \approx 144^\circ$.

B.3 Specific Predictions

Material	θ_0	α ($^\circ$ /GPa)	T_g	Condition
Silica (SiO ₂)	144 $^\circ$	~ 0	$\sim 1200^\circ\text{C}$	Already glass-like
Polyethylene	180 $^\circ$	N/A	-120°C	$T < -120^\circ\text{C}$
CO ₂ (solid)	180 $^\circ$	~ 10	N/A	$P > 5$ GPa
Hydrogen (Jupiter)	120 $^\circ$	~ 5	N/A	$P > 100$ GPa
Water (ice)	109.5 $^\circ$	~ 3	-140°C	$T < -140^\circ\text{C}$

B.4 The General Prediction

Any material with an average bond angle $\theta > 144^\circ$, or with an angle distribution width $\sigma_\theta > 20^\circ$, will become glass-like under the condition

$$P > \frac{\theta_{\text{glass}} - \theta_0}{\alpha} \quad \text{or} \quad T < T_g.$$

B.5 Philosophical Conclusion

*Glass is not a unique material — it is a **phase-state** that any material can enter, provided its bond angles approach the 180 $^\circ$ limit. Pressure compresses angles; temperature broadens them. The glass transition is not a chemical event — it is a phase-catastrophe, a collapse of phase-coherence when the network can no longer sustain its geometry.*

The Anti-Phase Catastrophe is universal. It applies to all materials — at the right temperature and pressure.

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